continuous observation in conflict zones, enabling the identification of targets, monitoring of enemy movements, and post-strike assessments.

- *Precision strike operations*: Armed drones introduced the ability to deliver laser-guided munitions such as AGM-114 Hellfire missiles, allowing for accurate engagement with minimal collateral damage. This capability reshaped counterterrorism operations in regions like Afghanistan, Iraq, and Pakistan.
- *Electronic warfare and communication relay*: Third-generation UAVs carried electronic warfare payloads to disrupt enemy radar and communication systems, while others served as airborne data relays to extend communication coverage in remote or contested areas.
- *Maritime and border surveillance*: Long-endurance drones such as the Heron TP and Global Hawk were used for maritime patrols, anti-piracy missions, and coastal defense, offering extended coverage and real-time situational awareness.
- *Target acquisition and battle damage assessment*: UAVs provided critical imaging and telemetry data used in guiding artillery, coordinating airstrikes, and evaluating strike effectiveness in near real-time.

4.4.2 Scientific and environmental applications

The robust endurance and precision of third-generation UAVs made them valuable assets for scientific research and environmental monitoring on a global scale.

- *Climate and atmospheric studies*: UAVs like NASA's Global Hawk conducted high-altitude missions to study weather systems, greenhouse gases, and atmospheric composition, collecting data beyond the reach of manned aircraft.